Kernix 0.11.1 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

35 workloads

123 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

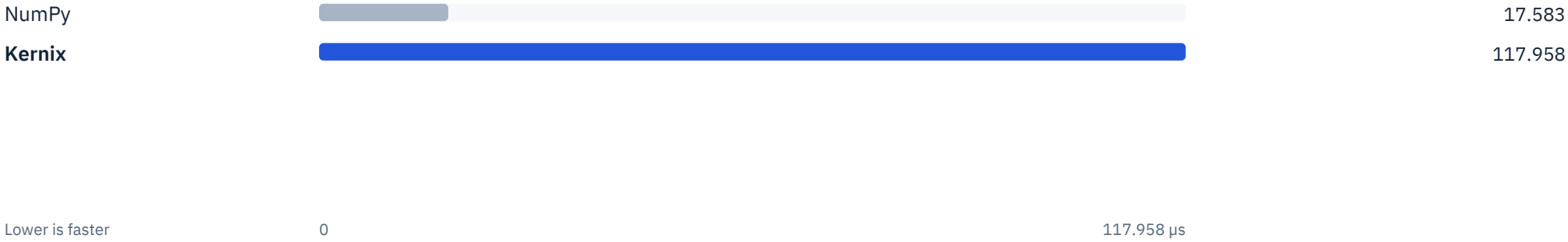
Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Independent matrix operands

Per-call median

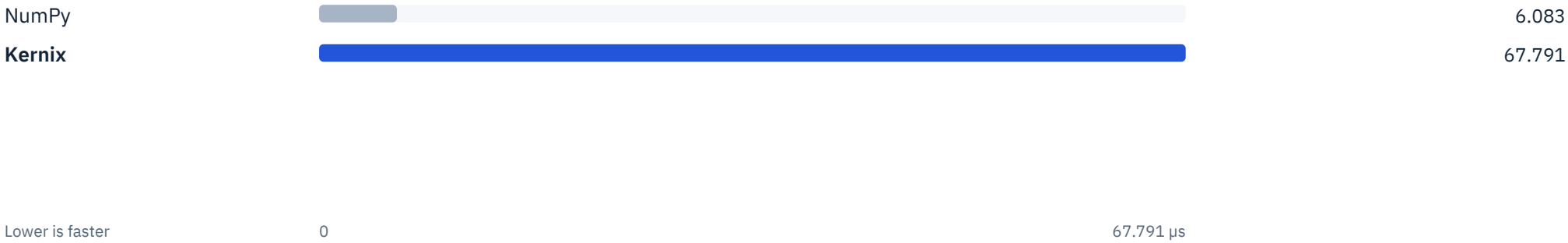
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



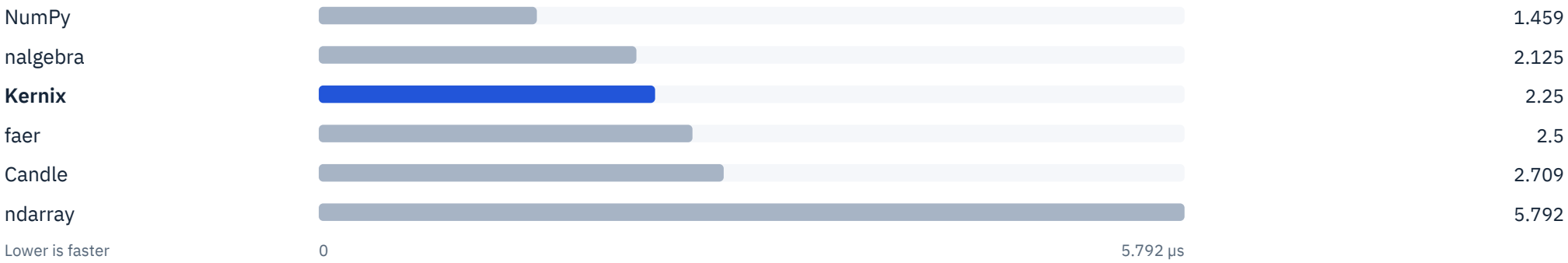
Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



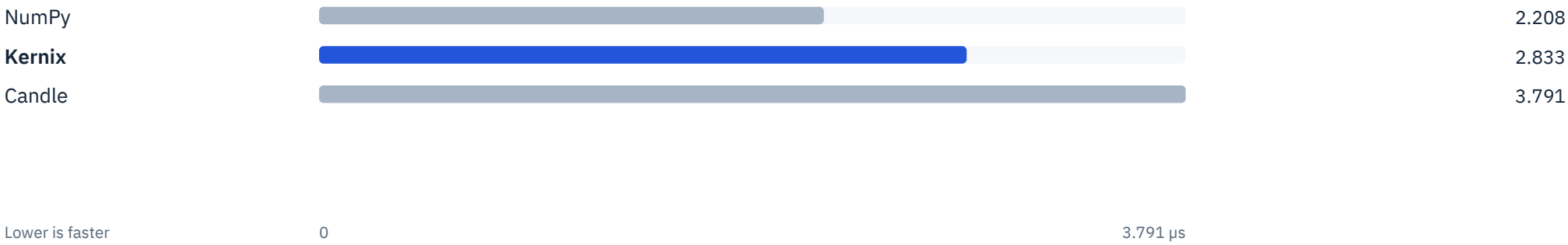
Kernix 0.11.1 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

Per-call median

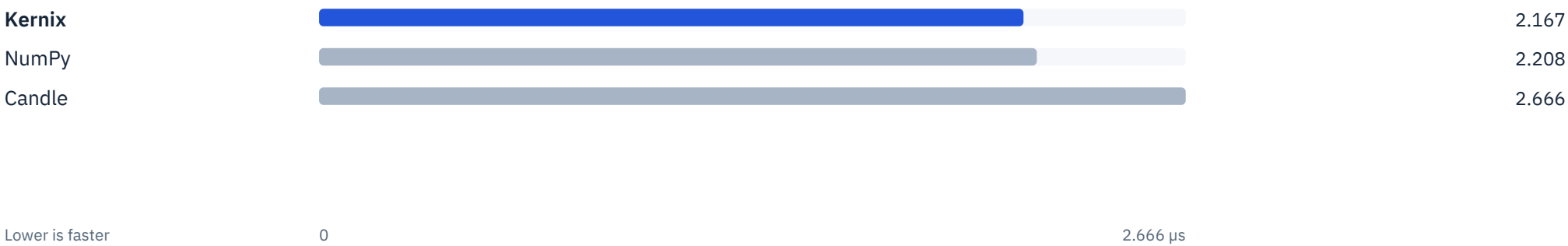
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.11.1 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.1 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



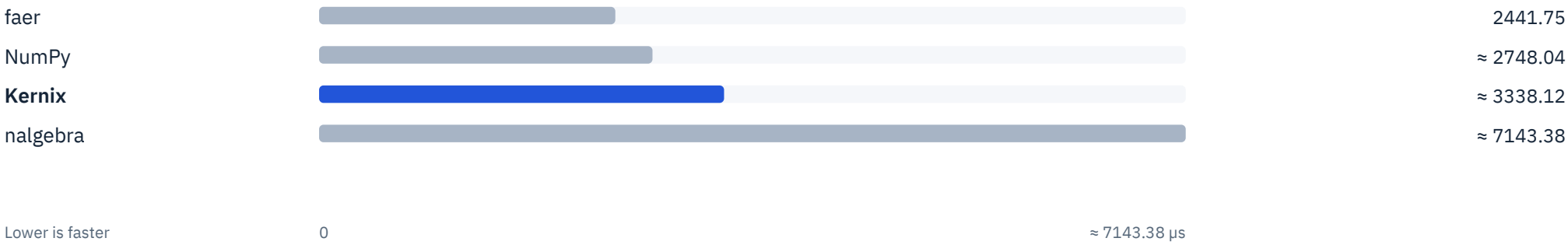
Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

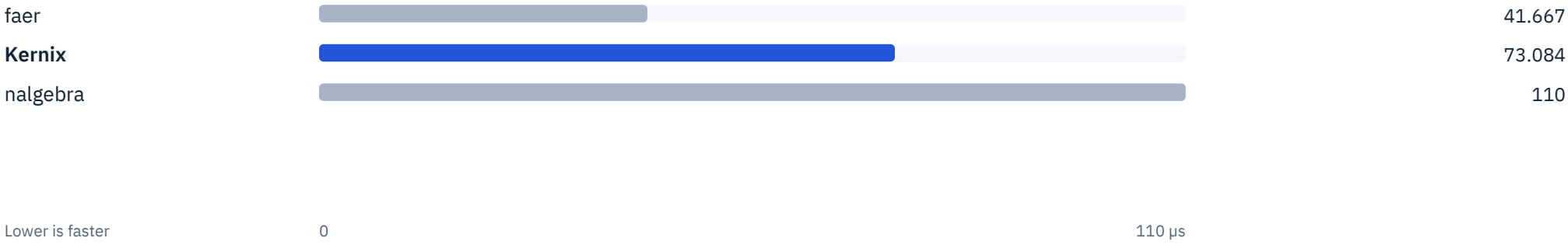
Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

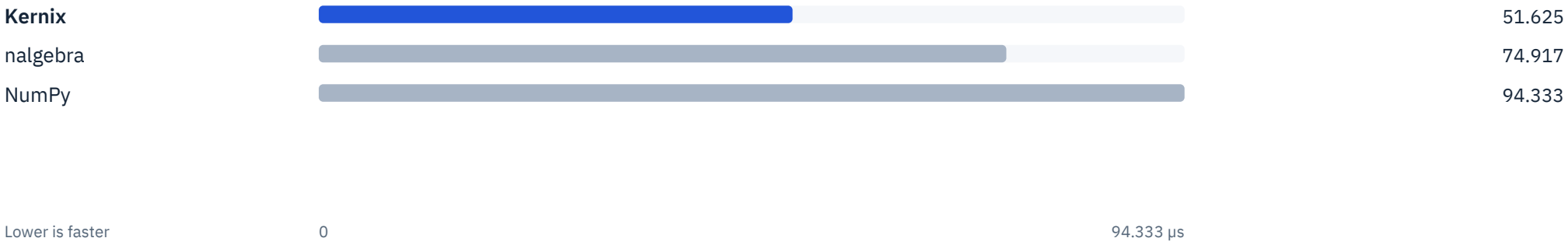
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Thin QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



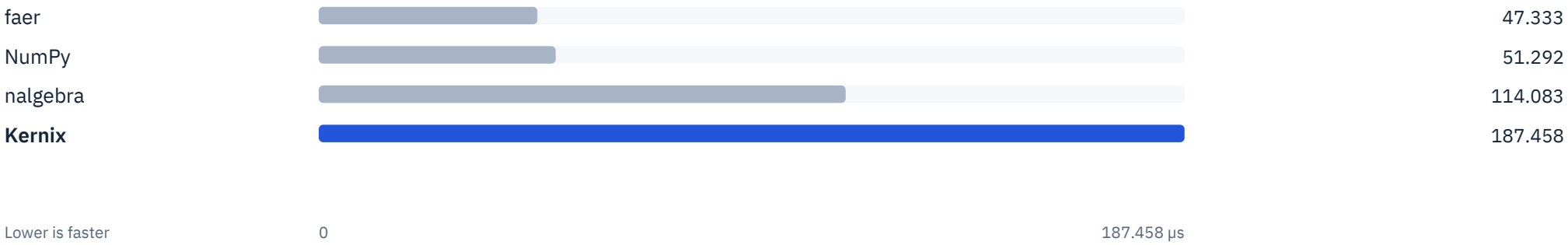
Kernix 0.11.1 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Shared-variable backward

Per-call median

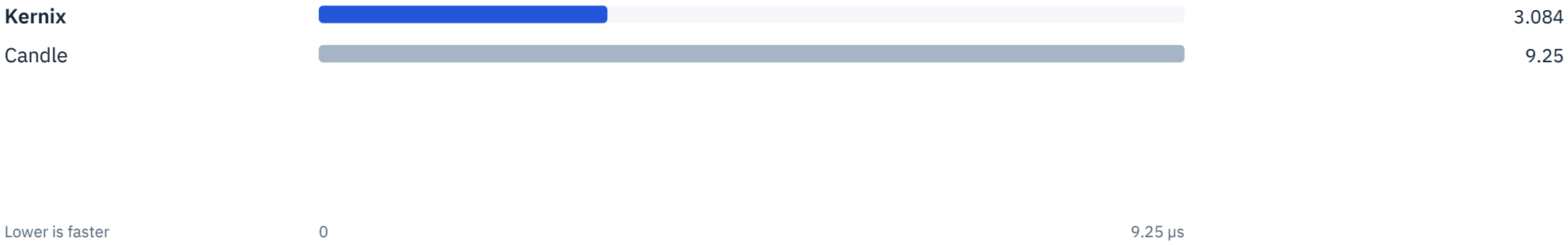
sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.11.1 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.1 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

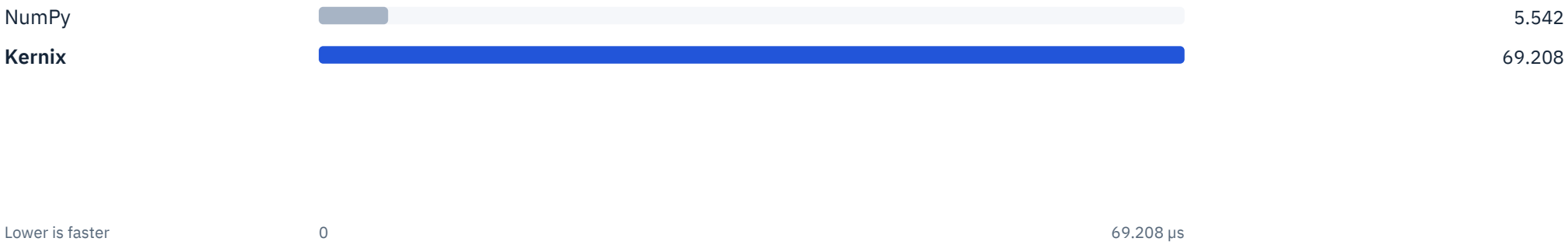


Kernix 0.11.1 / Tensor contractions

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output



Kernix 0.11.1 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output



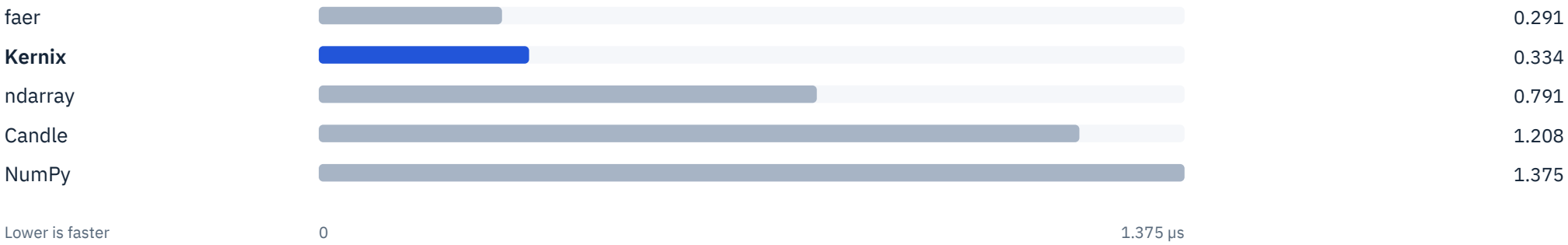
Kernix 0.11.1 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Create zeros

64 × 64 · f32 · 1 worker · allocated output

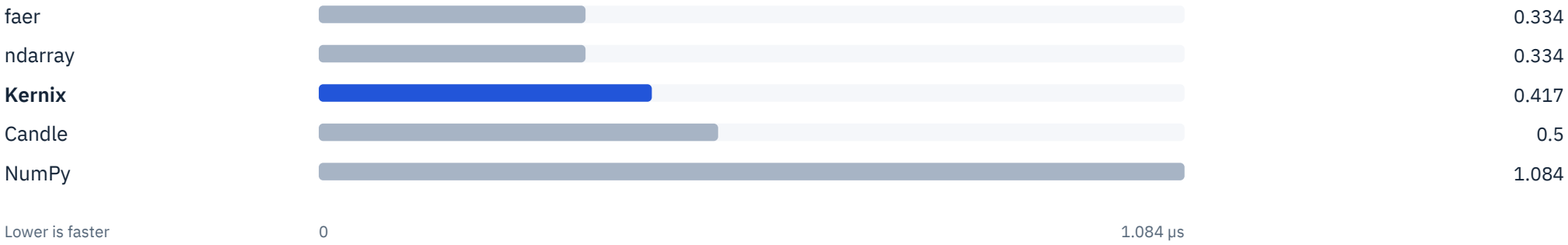
Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9851747264384045	0.7713404182379157 – 1.0742222222222222

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0052673086097617	0.9046403956437102 – 1.0933024369821536

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9760507399577167	0.8449660678642713 – 1.011104761904762

Matrix multiplication

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9985502367201311	0.9208476303739065 – 1.0949789086950605

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9856326348566568	0.8686482314018108 – 1.0417187028280885

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9230769230769231	0.7752203721841332 – 1.2563076923076921

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9936854234101746	0.9442126930923929 – 1.052763970607238

Independent matrix operands

128 × 128 · 128 × 128 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9093212885529866	0.8770501260982082 – 0.9883292181069959

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.9583333333333334 – 1.076

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear



CURRENT / BASELINE	1	95% INTERVAL	0.8832 – 1.024

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9997944119201927	0.9500575602961868 – 1.0822704081632653

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0057055889993636	0.9257904545228178 – 1.1766056170546626

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Timing direction unclear



CURRENT / BASELINE	1.0711008243106563	95% INTERVAL	0.9638131954788399 – 1.282426526539142

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	0.9512175381610398	95% INTERVAL	0.5288740230405842 – 1.398729942304536

Version comparison / paired processes

Apple M1 (Virtual) · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9764277002242503	0.9081814845657666 – 1.4083494301625017

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0443037974683544	0.5116346525718706 – 1.2821479846572525

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	17583	17.583
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	117958	117.958
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	6083	6.083
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	67791	67.791
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2125	2.125
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	5792	5.792
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2709	2.709
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2500	2.5
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1459	1.459
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2250	2.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	13333	13.333
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3667	3.667
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	14750	14.75
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	15916.000000000002	15.916000000000002
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	14792	14.792
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	14125	14.125
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	116500	116.5
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	19500	19.5
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	115500	115.5
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	113542	113.542

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	114500	114.5
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	112084	112.084
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f64	18000	18
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	125250	125.25
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	934500	934.5
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	989959.0000000001	989.9590000000001
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1071208	1071.208
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f64	124458	124.458
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1005250.0000000001	1005.2500000000001
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	821583	821.583

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	6917	6.917
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6959	6.959
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	8292	8.292
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1958	1.958
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	7333	7.333
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	8167	8.167
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	55042	55.042
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	55334	55.334
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	5959	5.959
Matrix multiplication / Kernix · explicit Accelerate 128 × 128 · 128 × 128 · f32	6333	6.333

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	49458	49.458
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	63792	63.792
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	60792	60.792
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	34916	34.916
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	442083	442.083
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	416042	416.042
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	433917	433.917
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	391375	391.375
Matrix multiplication / Kernix · explicit Accelerate 256 × 256 · 256 × 256 · f32	35792	35.792
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	401458	401.458

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3791	3.791
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2208	2.208
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2833	2.833
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2666	2.666
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	2208	2.208
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	2167	2.167
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	26708	26.708
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	157750	157.75
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	605917	605.917
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	99541	99.541

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / nalgebra 512 × 512 · f64	7143375	7143.375
Cholesky factorization / NumPy 512 × 512 · f64	2748042	2748.042
Cholesky factorization / faer 512 × 512 · f64	2441750	2441.75
Cholesky factorization / Kernix 512 × 512 · f64	3338125	3338.125
Column-pivoted QR / nalgebra 96 × 32 · f64	110000	110
Column-pivoted QR / Kernix 96 × 32 · f64	73084	73.084
Column-pivoted QR / faer 96 × 32 · f64	41667	41.667
Column-pivoted QR / faer 256 × 64 · f64	359500	359.5
Column-pivoted QR / Kernix 256 × 64 · f64	727625	727.625
Column-pivoted QR / nalgebra 256 × 64 · f64	1057666	1057.666

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	74917	74.917
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	94333	94.333
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	51625	51.625
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	195000	195
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	186333	186.333
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	73000	73
Thin QR / NumPy 96 × 32 · f64	61166	61.166
Thin QR / Kernix 96 × 32 · f64	63625	63.625
Thin QR / nalgebra 96 × 32 · f64	42000	42
Thin QR / faer 96 × 32 · f64	51958	51.958

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 256 × 64 · f64	316084	316.084
Thin QR / Kernix 256 × 64 · f64	622667	622.667
Thin QR / nalgebra 256 × 64 · f64	395792	395.792
Thin QR / NumPy 256 × 64 · f64	400958	400.958
Cholesky factorization / faer 64 × 64 · f64	10833	10.833
Cholesky factorization / NumPy 64 × 64 · f64	16833	16.833
Cholesky factorization / nalgebra 64 × 64 · f64	20917	20.917
Cholesky factorization / Kernix 64 × 64 · f64	22292	22.292
Cholesky factorization / NumPy 128 × 128 · f64	51292	51.292
Cholesky factorization / nalgebra 128 × 128 · f64	114083	114.083

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / faer 128 × 128 · f64	47333	47.333
Cholesky factorization / Kernix 128 × 128 · f64	187458	187.458
Cholesky factorization / Kernix 256 × 256 · f64	759250	759.25
Cholesky factorization / NumPy 256 × 256 · f64	431625	431.625
Cholesky factorization / faer 256 × 256 · f64	269916	269.916
Cholesky factorization / nalgebra 256 × 256 · f64	1094583	1094.583
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4334	4.334
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	6042	6.042
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	9250	9.25
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3084	3.084

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model inference / Candle batched normalized/sliced MLP · f64	15916.000000000002	15.916000000000002
Model inference / Kernix batched normalized/sliced MLP · f64	2459	2.459
Model recording + forward / Kernix batched normalized/sliced MLP · f64	4792	4.792
Model recording + forward / Candle batched normalized/sliced MLP · f64	6958	6.958
Model backward / Kernix prepared graph → fresh gradients · f64	8917	8.917
Model backward / Candle prepared graph → fresh gradients · f64	23667	23.667
Parameter update / Kernix transactional SGD, fixed gradients · f64	750	0.75
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	69208	69.208
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	5542	5.542
Row softmax / ndarray 256 × 64 · f64	67500	67.5

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Row softmax / Candle 256 × 64 · f64	82417	82.417
Row softmax / Kernix 256 × 64 · f64	92042	92.042
Row softmax / NumPy 256 × 64 · f64	89166	89.166
Create zeros / faer 64 × 64 · f32	291	0.291
Create zeros / Candle 64 × 64 · f32	1208	1.208
Create zeros / NumPy 64 × 64 · f32	1375	1.375
Create zeros / Kernix 64 × 64 · f32	334	0.334
Create zeros / ndarray 64 × 64 · f32	791	0.791
Create ones / ndarray 64 × 64 · f32	334	0.334
Create ones / NumPy 64 × 64 · f32	1084	1.084

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / Candle 64 × 64 · f32	500	0.5
Create ones / Kernix 64 × 64 · f32	417	0.417
Create ones / faer 64 × 64 · f32	334	0.334